

Applications of Bend and Break

Revised excerpt: original pp. 136–145

Conventions. The original ground field k is algebraically closed of characteristic zero. The symbols X, C denote the original varieties over k ; X_p, C_p (or X_P, C_P) denote geometric fibres of their models in positive characteristic. Here a Fano variety is smooth, projective, and of positive dimension. We retain the notation f_*C for the pushforward 1-cycle, including its mapping degree. The numbered results cited from the notes keep their original numbering.

Theorem 2.3.6 ([Mor82]). *For every point x of a Fano variety X , there is a rational curve D through x such that*

$$0 < -(D \cdot K_X) \leq \dim X + 1.$$

Proof. This is done in several steps. Write $n = \dim X$.

Step 1: To find this rational curve, we intend to use the bend-and-break technique. Thus we have to find a morphism $f : C \rightarrow X$ which we will be able to deform, then we bend these deformations together over a fixed point, and break them into some rational curves.

Pick an integral projective curve $C' \subseteq X$ passing through x , and let $f : C \rightarrow C' \hookrightarrow X$ be its normalization. Then C is a smooth projective curve. Pick $c_0 \in C$ such that $f(c_0) = x$, which is the target point we are bending.

Step 2: According to the formula in Section 2.1E,

$$\dim_{[f]} \text{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot n.$$

Thus whenever the quantity $-(K_X \cdot f_*C) - g(C)n$ is positive, there must be a non-trivial one-dimensional family of deformations of the map $f : C \rightarrow X$ keeping the image of c_0 fixed. So using Bend and Break I, we obtain a rational curve in X through x .

Step 3: Now we show how to get the positive quantity

$$-(K_X \cdot f_*C) - g(C) \cdot n > 0.$$

To do this, we need to get $-(K_X \cdot f_*C)$ to be big enough. We consider three cases:

(1) $g(C) = 0$.

Then automatically $-(K_X \cdot f_*C) > 0$, since $-K_X$ is ample. Actually, by Lüroth's theorem, the image of C is already a rational curve through x .

(2) $g(C) = 1$.

For the same reason we have $-(K_X \cdot f_*C) > 0$, but as we want it to be big enough, we apply some finite endomorphism of C fixing the point c_0 .

In particular, take c_0 as the origin of the elliptic curve C . Via the injective ring homomorphism [Har77, IV, 4.10]

$$\mathbb{Z} \longrightarrow \text{End}(C, c_0),$$

we can pick the multiplication-by- r endomorphism $[r]_C$ and compose it with f . Then

$$-(K_X \cdot (f \circ [r]_C)_* C) - n = -r^2(K_X \cdot f_* C) - n,$$

where the equality follows from $\deg[r]_C = r^2$ [Har77, IV, 4.17]. This is positive for $r \gg 0$, and $(f \circ [r]_C)(c_0) = x$. Thus a multiple of the original cycle admits deformations with the image of one point fixed.

$$(C, c_0) \xrightarrow[\deg=r^2]{[r]_C} (C, c_0) \xrightarrow{f} (X, x).$$

Figure 2.9. The endomorphism $[r]_C$ raises the mapping degree while preserving the genus and the marked point.

(3) $g(C) \geq 2$.

By the Hurwitz formula for a separable endomorphism of degree d , we have

$$2g(C) - 2 = d \cdot (2g(C) - 2) + \deg R_{\text{ram}}.$$

If $d > 1$, the right hand side is strictly larger than the left, since $\deg R_{\text{ram}} \geq 0$. So a curve of genus at least 2 has no separable endomorphism of degree greater than 1. In characteristic zero, every finite endomorphism is separable.

Thus we can try to consider another smooth projective curve $h : C'' \longrightarrow C$ and deform $f \circ h$. But the problem here is that even though

$$((f \circ h)_* C'' \cdot K_X) = \deg h \cdot (f_* C \cdot K_X),$$

the genus also changes. For example, if C'' is a connected r -sheeted unbranched cover of C , then by the Hurwitz formula again,

$$2g(C'') - 2 = r \cdot (2g(C) - 2).$$

Hence

$$\begin{aligned} & -(K_X \cdot (f \circ h)_* C'') - g(C'') \cdot n \\ &= -r((K_X \cdot f_* C) + g(C) \cdot n) + (r - 1)n. \end{aligned}$$

This does not necessarily become positive even by making r large enough.

Step 4: Thus we are only in trouble in the case $g(C) > 1$, because in characteristic zero C does not admit an endomorphism of high degree. To overcome this, we will first reduce the data to positive characteristic. In the situation of Step 5, write C_p for the smooth projective reduction of C over $\overline{\mathbb{F}}_p$, defined over $\mathbb{F}_p$. Raising its homogeneous coordinates to the p -th power gives an $\overline{\mathbb{F}}_p$ -endomorphism

$$F_p : C_p \longrightarrow C_p, \quad \deg F_p = p.$$

Its iterates $F_p^m : C_p \longrightarrow C_p$ have degree p^m , while the domain remains the same curve C_p and hence has the same genus. Thus Frobenius will be applied to the reduction C_p , not to the original characteristic-zero curve C .

Next we construct these reductions. Step 6 treats the general case, where the residue field may be $\mathbb{F}_q$ and we use the q -power Frobenius on C_P .

Step 5: First suppose that X, C , the morphism f , and the point c_0 are all defined over $\mathbb{Q}$. After clearing denominators, take equations with integral coefficients

$$\begin{aligned} h_1(x_0, \dots, x_N), \dots, h_a(x_0, \dots, x_N) & \text{ for } X, \\ c_1(y_0, \dots, y_M), \dots, c_b(y_0, \dots, y_M) & \text{ for } C. \end{aligned}$$

We also spread out the graph of f and the marked point c_0 . After inverting finitely many primes, these data give models

$$\mathcal{X} \longrightarrow U, \quad \mathcal{C} \longrightarrow U, \quad f_U : \mathcal{C} \longrightarrow \mathcal{X}, \quad U = \operatorname{Spec} \mathbb{Z}[1/N_0],$$

together with a section $c_{0,U} : U \longrightarrow \mathcal{C}$. Set $x_U = f_U \circ c_{0,U}$. We may shrink U so that $\mathcal{X} \longrightarrow U$ and $\mathcal{C} \longrightarrow U$ are smooth and projective with geometrically integral fibres, of relative dimensions n and 1 respectively.

The sheaves

$$\mathcal{O}_{\mathcal{C}}, \quad f_U^* \omega_{\mathcal{X}/U}, \quad f_U^* T_{\mathcal{X}/U}$$

are vector bundles on $\mathcal{C}$ and are flat over U . Since $\mathcal{C} \longrightarrow U$ is proper, their fibrewise Euler characteristics are locally constant. For instance, writing $f_p : C_p \longrightarrow X_p$ for the geometric fibre over (p) ,

$$(K_{X_p} \cdot (f_p)_* C_p) = \deg f_p^* \omega_{X_p} = \chi(f_p^* \omega_{X_p}) - \chi(\mathcal{O}_{C_p}).$$

As U is connected, the quantities

$$(K_{X_p} \cdot (f_p)_* C_p), \quad g(C_p), \quad \chi(f_p^* T_{X_p})$$

are constant, and agree with their values on the original curve.

Moreover, ω_X^{-1} extends to $\omega_{\mathcal{X}/U}^{-1}$. By openness of ampleness on the fibres of a proper morphism, we may shrink U again so that $-K_{X_p}$ is ample for every geometric fibre. Hence

$$A := -(K_{X_p} \cdot (f_p)_* C_p) = -(K_X \cdot f_* C) > 0.$$

The geometric fibres C_p and X_p are defined over $\mathbb{F}_p$. Raising the homogeneous coordinates of C_p to the p -th power gives the Frobenius endomorphism F_p , a finite morphism of degree p . Since the section $c_{0,U}$ was included in the data, $c_{0,p}$ is $\mathbb{F}_p$ -rational and $F_p^m(c_{0,p}) = c_{0,p}$. Therefore the base-pointed deformation space of the composite morphism

$$C_p \xrightarrow{F_p^m} C_p \xrightarrow{f_p} X_p$$

has dimension bounded below by

$$\begin{aligned} \dim_{[f_p \circ F_p^m]} \text{Mor}(C_p, X_p; (f_p \circ F_p^m)|_{\{c_{0,p}\}}) \\ \geq -p^m(K_{X_p} \cdot (f_p)_*C_p) - g(C_p) \cdot n \\ = p^m A - g(C)n. \end{aligned}$$

This is positive for $m \gg 0$. Then as in Step 2, we produce a rational curve A_p on X_p through x_p for all primes represented in U .

Step 6: Suppose now that we are in the general case. The coefficients of the h_j defining X , the c_j defining C , the equations defining the graph $\Gamma(f)$, and the coordinates of c_0 need not be rational. These finitely many coefficients lie in a finitely generated $\mathbb{Z}$ -subalgebra $R \subseteq k$. After adjoining the necessary coefficients and localizing R , we obtain models and sections as in Step 5 over a non-empty affine open subset $U \subseteq \text{Spec } R$. We may again arrange smoothness, projectivity, geometric integrality, constancy of the above numerical quantities, and ampleness of the anticanonical bundles on the geometric fibres.

Now let $P \in U$ be a closed point. By Lemma 2.3.3, its residue field $\kappa(P)$ is finite. Set $q = |\kappa(P)|$, and denote the geometric fibres by C_P and X_P . The q -power Frobenius endomorphism $F_q : C_P \rightarrow C_P$ fixes $c_{0,P}$, and its m -th iterate has degree q^m . Thus

$$\begin{aligned} \dim_{[f_P \circ F_q^m]} \text{Mor}(C_P, X_P; (f_P \circ F_q^m)|_{\{c_{0,P}\}}) \\ \geq -q^m(K_{X_P} \cdot (f_P)_*C_P) - g(C_P) \cdot n > 0 \end{aligned}$$

for $m \gg 0$. This gives a rational curve A_P through x_P for every closed point $P \in U$.

Step 7: In this step, we wish to obtain the required inequality, so we wish to replace A_P with another rational curve B_P through x_P such that

$$0 < -(K_{X_P} \cdot B_P) \leq n + 1.$$

To do this, notice that if $-(K_{X_P} \cdot A_P) > n + 1$, then we can take the normalization morphism

$$\varphi : \mathbb{P}^1_{\kappa(P)} \rightarrow A_P \hookrightarrow X_P.$$

In particular, $\varphi_*\mathbb{P}^1 = A_P$ as 1-cycles. Choose $q_0 \in \mathbb{P}^1$ with $\varphi(q_0) = x_P$, and choose $q_1 \neq q_0$. According to Section 2.1E, with these two point images fixed, we have

$$\begin{aligned} \dim_{[\varphi]} \text{Mor}(\mathbb{P}^1, X_P; \varphi|_{\{q_0, q_1\}}) &\geq -(K_{X_P} \cdot \varphi_*\mathbb{P}^1) - n \\ &= -(K_{X_P} \cdot A_P) - n > 1. \end{aligned}$$

So this deformation space has dimension at least two.

But notice that φ is birational onto its image. Parametrizations with the same image differ by an automorphism of its normalization. Keeping the two point images fixed allows only finitely many choices of their preimages on the normalization; for each such choice, the remaining automorphisms form a one-dimensional family. Hence the fixed-image locus near $[\varphi]$ has dimension at most one, and we can choose a one-dimensional family in which the image of $\mathbb{P}^1$ actually moves.

By Bend and Break II, we obtain

$$A_P = \varphi_*\mathbb{P}^1 \sim_{\text{alg}} \sum_i a_i C_i, \quad a_i \in \mathbb{Z}_{>0}, \quad x_P \in \bigcup_i C_i,$$

where the C_i are integral rational curves and the effective cycle on the right is non-integral: it is reducible or a multiple rational curve. Since $-K_{X_P}$ is ample, each component satisfies

$$0 < -(K_{X_P} \cdot C_i) < -(K_{X_P} \cdot A_P).$$

This strict inequality also holds when the right hand side is a single multiple curve aC with $a > 1$.

Choose a component passing through x_P , take its normalization, and continue this process of deformation and decomposition. As long as the anticanonical degree is greater than $n + 1$, we find a rational curve through x_P of strictly smaller positive integral degree. This process must terminate in finitely many steps, giving our desired rational curve B_P with

$$x_P \in B_P, \quad 0 < -(K_{X_P} \cdot B_P) \leq n + 1.$$

Step 8: In this step, we must conclude the existence of a rational curve on the original Fano variety X , in particular in characteristic zero, from the existence of bounded degree rational curves B_P for every closed point $P \in U$.

Choose a positive integer ℓ such that $-\ell K_X$ is very ample, and fix the corresponding embedding $X \hookrightarrow \mathbb{P}_k^N$. By enlarging R and shrinking U if necessary, spread out this fixed embedding to

$$\mathcal{X} \hookrightarrow \mathbb{P}_U^N, \quad \mathcal{O}_{\mathcal{X}}(1) \simeq \omega_{\mathcal{X}/U}^{-\ell}.$$

All the preceding arguments apply to these models. Since a closed immersion remains a closed immersion after base change, this gives embeddings of all the geometric fibres into a fixed projective space. The degree of B_P under this embedding is

$$-\ell(K_{X_P} \cdot B_P) \leq \ell(n+1).$$

Set $D_0 = \ell(n+1)$. For $1 \leq e \leq D_0$, consider the relative space of morphisms of degree e , measured by this embedding, with the image of $0 = [0 : 1] \in \mathbb{P}^1$ fixed at x_U . Their disjoint union is

$$\mathcal{M} = \coprod_{e=1}^{D_0} \text{Mor}_e(\mathbb{P}_U^1, \mathcal{X}; 0 \mapsto x_U).$$

This is a scheme of finite type over U . On each geometric fibre, Mor_e parametrizes tuples of homogeneous polynomials of degree e , up to simultaneous non-zero scaling, with no common zero, satisfying the equations of the target and the prescribed point condition. Thus these are actual morphisms, and $e \geq 1$ excludes constant maps. We allow all degrees between 1 and D_0 instead of multiplying coordinates by a common factor.

For every closed point $P \in U$, the normalization of B_P , after reparametrizing so that 0 maps to x_P , gives a geometric point of $\mathcal{M}_P$. By Chevalley's theorem, the image of $\mathcal{M} \rightarrow U$ is constructible. Since U is an integral scheme of finite type over $\mathbb{Z}$, its closed points are dense. This constructible image contains every closed point, hence contains the generic point η of U .

Therefore $\mathcal{M}_\eta$ is non-empty. Write $K = \text{Frac } R \subseteq k$. A closed point of $\mathcal{M}_\eta$ has residue field finite over K , and this field embeds into the original algebraically closed field k over K . After base change, we obtain a non-constant morphism

$$g : \mathbb{P}_k^1 \rightarrow X, \quad g(0) = x, \quad 1 \leq \deg g^* \mathcal{O}_X(1) \leq D_0.$$

Here we return to the original field k containing R ; no embedding of $\text{Frac } R$ into an arbitrary algebraically closed field is required.

Step 9: Let $D = g(\mathbb{P}^1)_{\text{red}}$. By Lüroth's theorem, D is a rational curve, and $x \in D$. If δ is the degree of g onto its image, then $g_* \mathbb{P}^1 = \delta D$, so

$$\ell \delta (-(K_X \cdot D)) = \deg g^* \mathcal{O}_X(1) \leq \ell(n+1).$$

Hence $-(K_X \cdot D) \leq n+1$. This number is strictly positive since $-K_X$ is ample. Thus D satisfies the required inequality. $\square$

2.3B Finding Rational Curves when K_X is not Nef

Now let us weaken our hypothesis on the smooth projective variety X . We no longer require $-K_X$ to be ample; we only assume that K_X is not nef, which means that there is an integral curve $C' \subseteq X$ such that

$$-(K_X \cdot C') > 0.$$

This imposes no nefness assumption on $-K_X$. We also fix an ample Cartier divisor H on X .

Theorem 2.3.7 ([Mor82]). *Let X be a smooth projective variety and H an ample Cartier divisor on X . Assume that there is an integral curve $C' \subseteq X$ such that $-(C' \cdot K_X) > 0$. Then there is a rational curve $E \subseteq X$ such that*

$$0 < -(E \cdot K_X) \leq \dim X + 1, \quad \frac{-(E \cdot K_X)}{E \cdot H} \geq \frac{-(C' \cdot K_X)}{C' \cdot H}.$$

Proof. For finding rational curves, we can use a similar technique as before, passing to fields of positive characteristic to ensure that the dimension of the deformation space is large enough. Write $n = \dim X$.

Step 1: Let $f : C \rightarrow C' \hookrightarrow X$ be the normalization of C' . It is birational onto its image and $f_*C = C'$ as 1-cycles. Choose $c_0 \in C$. If

$$\dim_{[f]} \text{Mor}(C, X; f|_{\{c_0\}}) \geq -(K_X \cdot f_*C) - g(C) \cdot n > 0,$$

then f can deform with the image of c_0 fixed. As before, using Bend and Break I, we obtain

$$f_*C \sim_{\text{alg}} f'_*C + Z,$$

where $f' : C \rightarrow X$ may be constant and Z is a non-zero effective rational 1-cycle. We keep the pushforward cycles here, since the remaining morphism need not be birational onto its image.

Step 2 (Repeating bend and break): In order to achieve the above lower bound on the deformation space, we pass to positive characteristic as in Steps 5–6 of Theorem 2.3.6. Spread out X, C, f, c_0 , and also H , over a non-empty affine integral scheme U of finite type over $\mathbb{Z}$. After shrinking U , take smooth projective models $\mathcal{X}, \mathcal{C}$, a morphism $f_U : \mathcal{C} \rightarrow \mathcal{X}$ extending f , and a relatively ample line bundle $\mathcal{H}$ extending $\mathcal{O}_X(H)$.

For a closed point $P \in U$, use the same notation $f_P : C_P \rightarrow X_P$ for the geometric fibres, and write H_P for the corresponding divisor class. Set

$$A = -(K_{X_P} \cdot (f_P)_*C_P), \quad B = H_P \cdot (f_P)_*C_P, \quad G = g(C_P)n.$$

These quantities are independent of P , with $A, B > 0$. In particular,

$$M := \frac{-(C' \cdot K_X)}{C' \cdot H} = \frac{-(K_{X_P} \cdot (f_P)_*C_P)}{H_P \cdot (f_P)_*C_P} = \frac{A}{B} > 0.$$

Here constancy follows from the same Euler characteristic argument as in Theorem 2.3.6, applied also to $f_U^* \mathcal{H}$.

Put $q = |\kappa(P)|$, and compose f_P with the m -th iterate of the q -power Frobenius $F_q : C_P \rightarrow C_P$. Writing $Q = q^m$, we get

$$\begin{aligned} \dim_{[f_P \circ F_q^m]} \text{Mor}(C_P, X_P; (f_P \circ F_q^m)|_{\{c_{0,P}\}}) \\ \geq -Q(K_{X_P} \cdot (f_P)_* C_P) - G = QA - G, \end{aligned}$$

which is positive for $m \gg 0$. The ratio M does not change after this composition, since both intersection numbers are multiplied by Q .

Set $f_{P,m}^0 = f_P \circ F_q^m$ and

$$Y_j = (f_{P,m}^j)_* C_P, \quad Y_0 = Q(f_P)_* C_P.$$

Whenever $-(K_{X_P} \cdot Y_{j-1}) > G$, we can deform the remaining morphism with the image of $c_{0,P}$ fixed and apply Bend and Break I again, without composing with Frobenius again. This gives

$$Y_{j-1} \sim_{\text{alg}} Y_j + W_j,$$

where W_j is a non-zero effective rational cycle. The image of the marked point is fixed for the deformation at each individual stage; we do not need it to remain the same between stages.

Recall that algebraic equivalence implies numerical equivalence [Ful98, 10.3]. Since H_P is ample, at each step

$$H_P \cdot Y_j = H_P \cdot Y_{j-1} - H_P \cdot W_j < H_P \cdot Y_{j-1}.$$

These are non-negative integers, so the process must stop after finitely many steps, with

$$Q(f_P)_* C_P \sim_{\text{alg}} Y_s + Z_s, \quad Z_s = \sum_{j=1}^s W_j, \quad -(K_{X_P} \cdot Y_s) \leq G.$$

This is our stopping condition; it does not assert that the remaining morphism has no deformations. The cycle Y_s may be zero.

Let

$$\begin{aligned} a &= -(K_{X_P} \cdot Y_s), & b &= -(K_{X_P} \cdot Z_s), \\ c &= H_P \cdot Y_s, & d &= H_P \cdot Z_s. \end{aligned}$$

Then

$$a \leq G, \quad c \geq 0, \quad a + b = QA, \quad c + d = QB.$$

For large m , it follows that $b \geq QA - G > 0$ and $0 < d \leq QB$. We only need the upper bound on a ; no lower bound on a is assumed.

Step 3: We claim the following. For any $\varepsilon > 0$, if $m \gg 0$, then there exists an integral component E_P of the rational cycle Z_s such that

$$\frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} > M - \varepsilon.$$

To see this, we use the following lemma.

Lemma 2.3.8. *If $c, d > 0$ and $a, b \in \mathbb{R}$, then*

$$\frac{a+b}{c+d} \leq \max \left\{ \frac{a}{c}, \frac{b}{d} \right\}.$$

Indeed, $(a+b)/(c+d)$ is the weighted average of a/c and b/d , with positive weights c and d . The same argument applies to any finite number of terms.

In our situation, for Q sufficiently large, we have

$$\frac{b}{d} \geq \frac{QA - G}{QB} = M - \frac{G}{QB} > M - \varepsilon.$$

This estimate also applies when $c = 0$; there is no need to form a/c . Write

$$Z_s = \sum_i r_i E_{P,i}, \quad r_i \in \mathbb{Z}_{>0},$$

and set $b_i = -(K_{X_P} \cdot E_{P,i})$ and $d_i = H_P \cdot E_{P,i} > 0$. By the weighted-average argument,

$$\max_i \left\{ \frac{b_i}{d_i} \right\} \geq \frac{\sum_i r_i b_i}{\sum_i r_i d_i} = \frac{b}{d} > M - \varepsilon.$$

Thus the required component E_P exists.

Step 4: From now on, take $0 < \varepsilon < M/2$. Suppose that

$$-(E_P \cdot K_{X_P}) > n + 1$$

for the rational curve E_P obtained above. Take its normalization $\varphi : \mathbb{P}^1 \rightarrow E_P \hookrightarrow X_P$ and fix the images of two distinct points $q_0, q_1 \in \mathbb{P}^1$. Then

$$\dim_{[\varphi]} \text{Mor}(\mathbb{P}^1, X_P; \varphi|_{\{q_0, q_1\}}) \geq -(E_P \cdot K_{X_P}) - n > 1.$$

As in Step 7 of Theorem 2.3.6, the fixed-image locus has dimension at most one. Hence we can move the image curve and apply Bend and Break II to obtain a non-integral effective rational cycle

$$E_P \sim_{\text{alg}} \sum_i a_i E'_{P,i}.$$

Using Lemma 2.3.8 with the multiplicities a_i , at least one component E'_P satisfies

$$\frac{-(E'_P \cdot K_{X_P})}{E'_P \cdot H_P} > M - \varepsilon. \quad (2.1)$$

Since $M - \varepsilon > 0$, this component has positive anticanonical degree. Also,

$$0 < H_P \cdot E'_P < H_P \cdot E_P,$$

because H_P is ample and the cycle is non-integral. This includes the case of a multiple rational curve.

If $-(E'_P \cdot K_{X_P}) > n + 1$, take its normalization and repeat. This process cannot continue infinitely, since at each step the positive integer $H_P \cdot E_P$ decreases. Eventually we arrive at a rational curve, again called E_P , such that

$$0 < -(E_P \cdot K_{X_P}) \leq n + 1, \quad \frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} > M - \varepsilon.$$

No nefness of $-K_{X_P}$ is needed here: positivity on the selected component follows from its ratio, and termination follows from H_P .

These inequalities imply

$$1 \leq E_P \cdot H_P < \frac{n + 1}{M - \varepsilon} < \frac{2(n + 1)}{M}.$$

Therefore, for all $0 < \varepsilon < M/2$ and all closed points P , the possible ratios belong to the same finite set

$$\mathcal{S} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}_{>0}, \quad a \leq n + 1, \quad b < \frac{2(n + 1)}{M} \right\}.$$

If $\mathcal{S}$ contains numbers below M , choose $\varepsilon > 0$ smaller than $M/2$ and smaller than the minimum of the positive numbers $M - s$ for $s \in \mathcal{S}$, $s < M$. If it contains no such number, any $0 < \varepsilon < M/2$ will do. The ratio $> M - \varepsilon$ must then be at least M . Consequently, for every closed point $P \in U$, we obtain a rational curve E_P with

$$0 < -(E_P \cdot K_{X_P}) \leq n + 1, \quad \frac{-(E_P \cdot K_{X_P})}{E_P \cdot H_P} \geq M.$$

This makes precise the argument of letting $\varepsilon \rightarrow 0$: we use finitely many numerical possibilities, without taking a limit of the curves themselves.

Furthermore, $H_P \cdot E_P \leq (n + 1)/M$, uniformly in P . We can now reason as in Step 8 of Theorem 2.3.6, keeping both intersection numbers in the parameter space.

Choose $\ell > 0$ such that ℓH is very ample, and spread out the corresponding fixed embedding, after enlarging R and shrinking U if necessary, with

$$\mathcal{O}_{\mathcal{X}}(1) \simeq \mathcal{H}^{\otimes \ell}.$$

The preceding argument applies to every closed fibre of the resulting model. For each of the finitely many pairs of integers

$$1 \leq a \leq n + 1, \quad b > 0, \quad \frac{a}{b} \geq M,$$

let $\mathcal{N}_{a,b}$ be the locus in $\text{Mor}_{\ell b}(\mathbb{P}_U^1, \mathcal{X})$ where

$$\deg g^* \omega_{\mathcal{X}/U}^{-1} = a, \quad \deg g^* \mathcal{H} = b.$$

The degrees of these pullback line bundles are locally constant in families of morphisms, so these loci are open and closed in their respective fixed-degree morphism spaces. Thus

$$\mathcal{N} = \coprod_{(a,b)} \mathcal{N}_{a,b}$$

is of finite type over U .

The normalization of each E_P gives a geometric point of $\mathcal{N}_P$. Chevalley's theorem therefore implies, as before, that $\mathcal{N}$ has a non-empty generic fibre. Base changing to the original algebraically closed field k , we obtain a non-constant morphism $g : \mathbb{P}_k^1 \rightarrow X$ with

$$a = \deg g^* \omega_X^{-1}, \quad b = \deg g^* \mathcal{O}_X(H), \quad 1 \leq a \leq n+1, \quad \frac{a}{b} \geq M.$$

Let $E = g(\mathbb{P}^1)_{\text{red}}$, which is rational by Lüroth's theorem, and let δ be the degree of g onto E . Then

$$-(K_X \cdot E) = \frac{a}{\delta}, \quad H \cdot E = \frac{b}{\delta}.$$

Hence

$$0 < -(K_X \cdot E) \leq n+1, \quad \frac{-(K_X \cdot E)}{H \cdot E} = \frac{a}{b} \geq M,$$

which is the required conclusion. $\square$

From this theorem we see the strength of the bend-and-break technique: Frobenius makes the deformation space large in positive characteristic, and bounded degree allows us to recover a rational curve over the original field.

References

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- [Har77] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977.
- [Ful98] W. Fulton, *Intersection Theory*, second edition, Springer, 1998.

Additional references for the revised passages. For the bounded-degree morphism space argument, see O. Debarre, *Bend and break*, pp. 24–25. For the two openness and constructibility facts, see the Stacks Project, Tag 0D3A (fibrewise ampleness) and Tag 054H (Chevalley's theorem).